

TRELLIS

by lemnisca
trellis.lemnisca.bio

The shortest path for your bioprocess *from plate to pilot to production*

Find the true potential of your process, establish robustness and advance
to production with confidence.

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The problem: You are optimizing a process you can only partially observe

From a limited experimental programme, teams must infer which process parameters drive performance, which interactions matter, where the true potential of the process lies, how broad the operating region is, and where the process may fail.

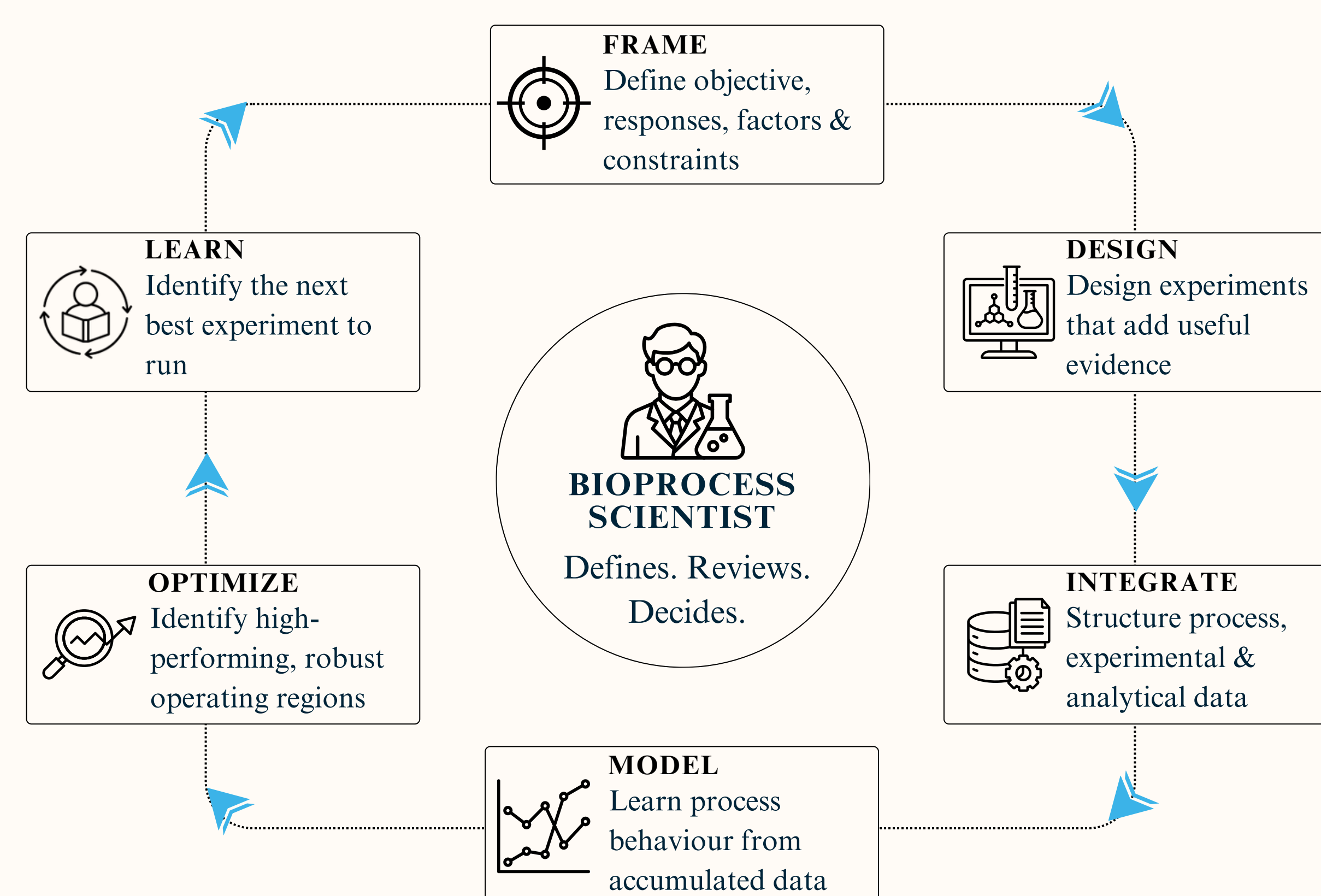
As the process moves from plate to pilot to production, the number of experiments falls while the stakes of each decision rise. Every experiment must therefore provide the most useful evidence for the next decision. Yet deciding what to test next, with only limited prior data to guide the choice, is daunting.



The resolution: Trellis makes a process more observable, with fewer experiments

Trellis combines accumulated experimental evidence with scientific knowledge to build an evolving model of the process from plate to pilot to production. It simulates conditions that cannot all be tested experimentally to reveal where the true potential of the process lies, how broad the operating region is, and where the process may fail.

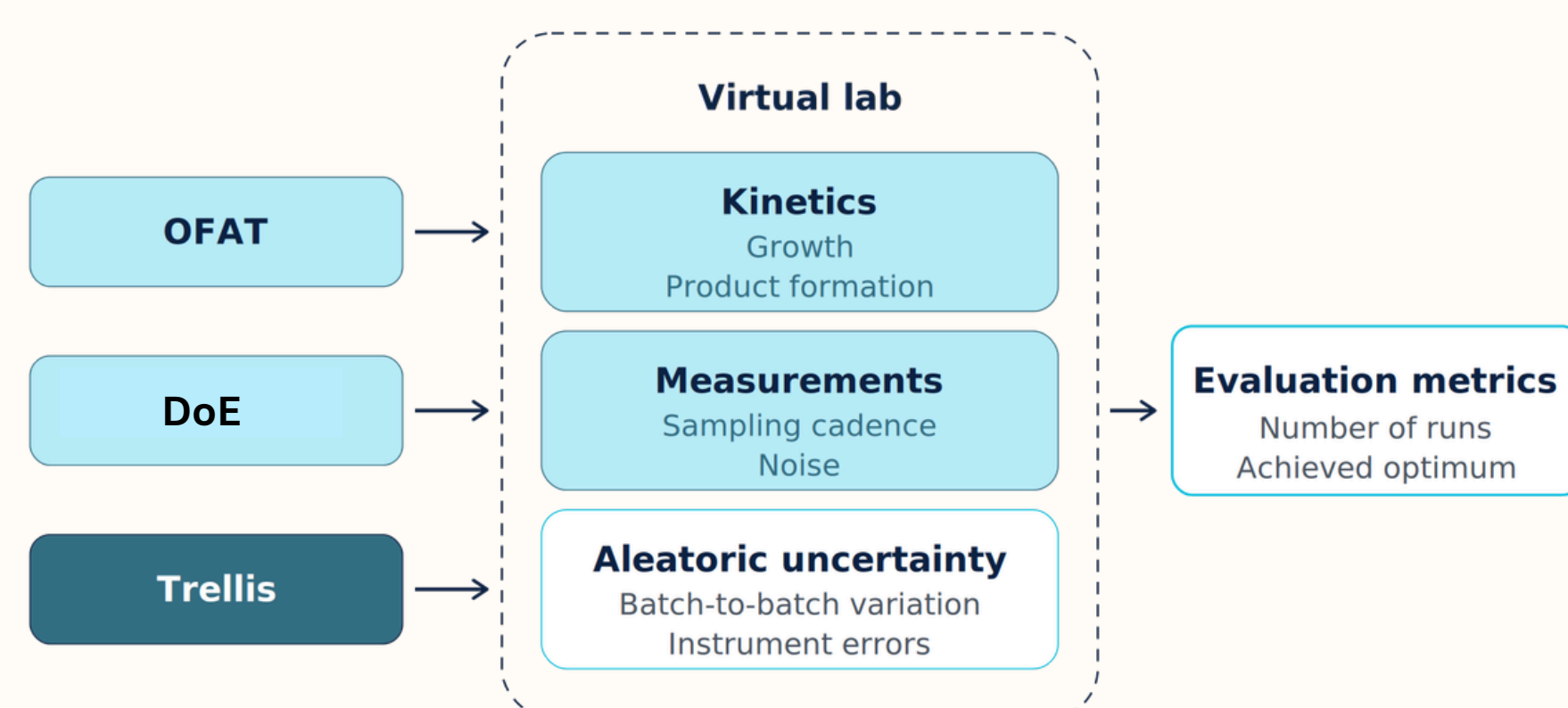
From this, Trellis recommends the experiment most likely to produce evidence that matters for the next decision. The bioprocess scientist drives the entire loop: defining the objective, factors and constraints, reviewing what the model has learned, and deciding whether the model needs to be refined before the next experiment is run.



The benchmark: Three experiment-design strategies tested on the same process in a virtual lab

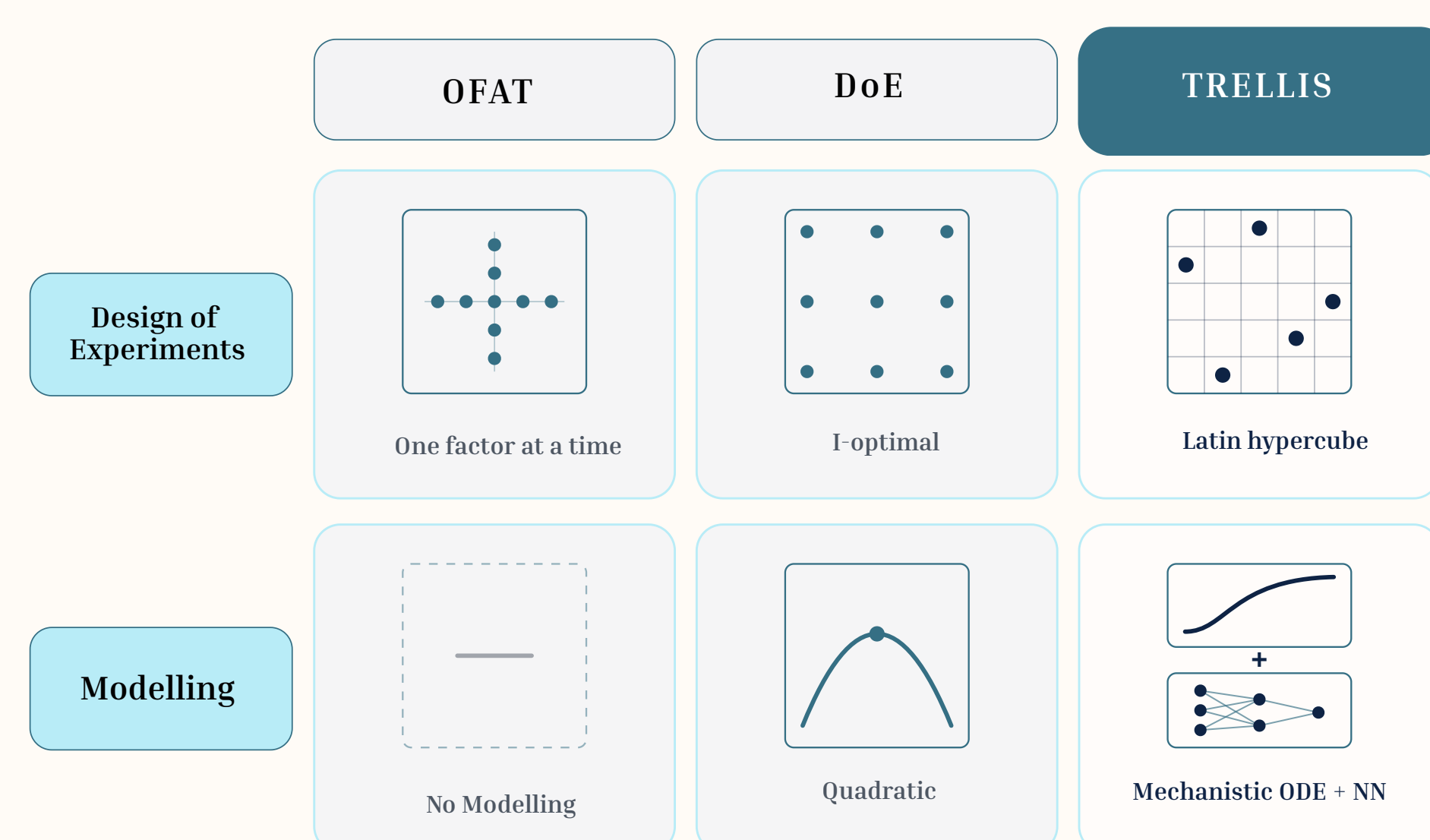
Benchmark results: Trellis significantly outperforms DoE and OFAT

Validating Trellis in a virtual lab



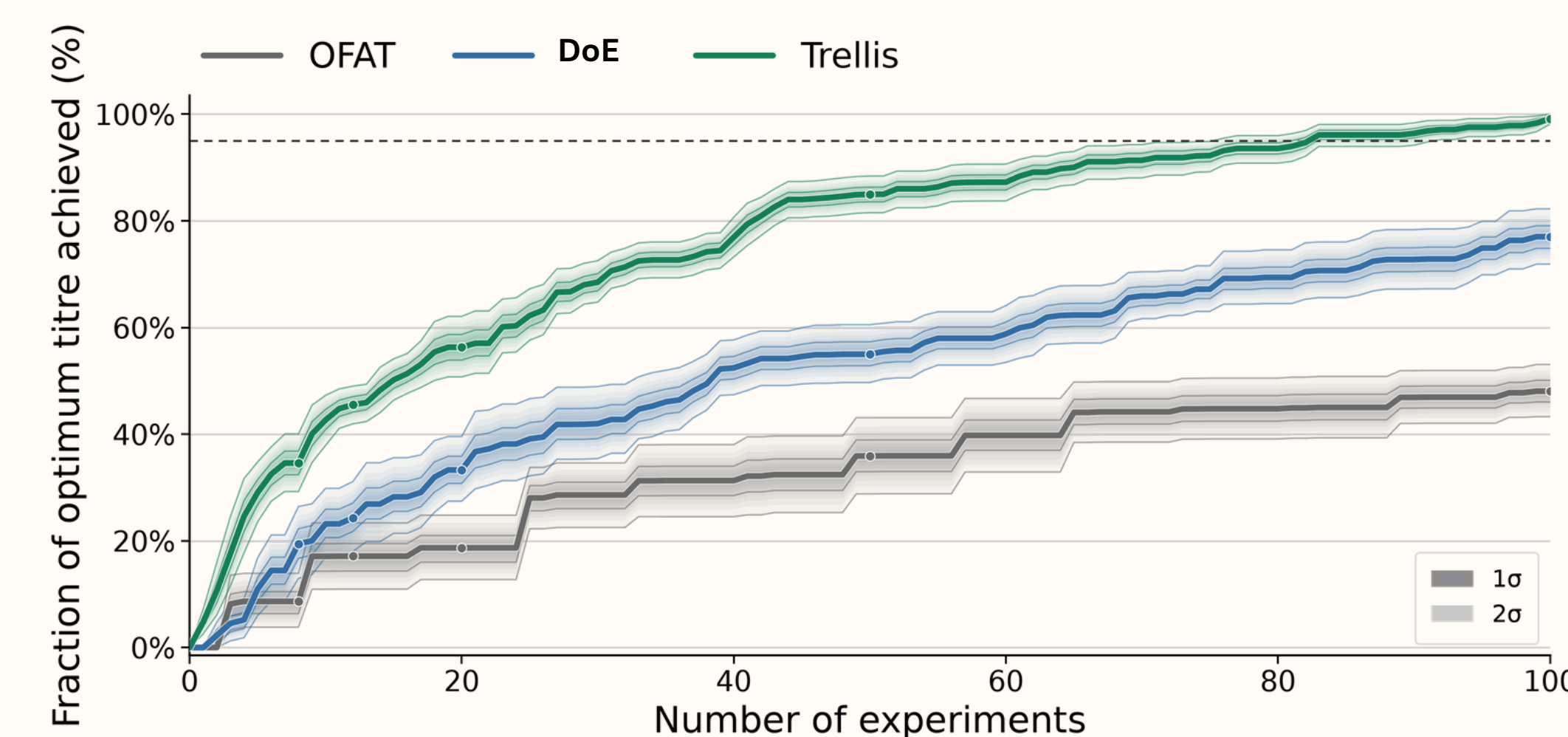
- 12 factors considered in the experiment design, including growth and production temperature, shift day, feed start day, daily feed volume, and seven feed concentrations.
- Multiple measurements were noted, including VCD, viability, titre, temperature, concentrations of glucose, glutamate, lactate, ammonia, and osmolality.
- **Objective was to maximize titre.**

Distinction between strategies



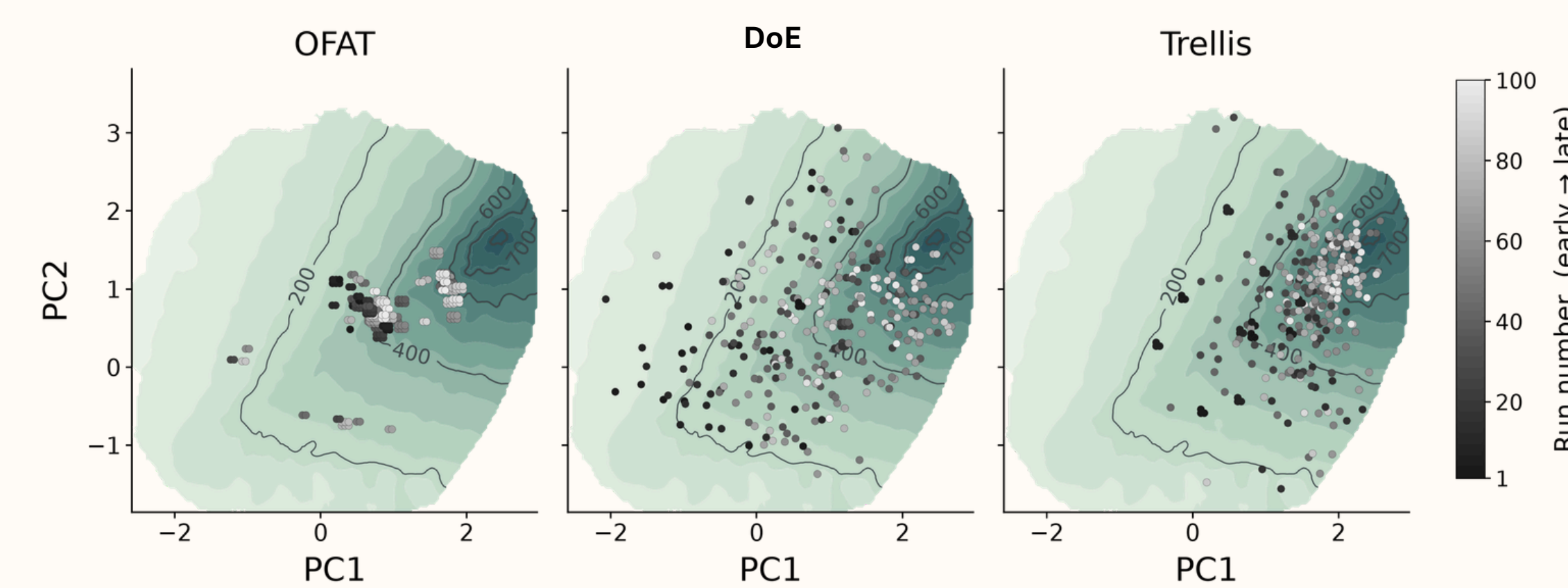
Optimization of the hybrid model in Trellis is performed by balancing maximization of titre and minimization of uncertainty through a Bayesian acquisition function.

Trellis gets closest to the known maximum titer, faster



Trellis improves titre faster and to a greater extent compared to simpler statistical models

Trellis samples the design space intelligently



- OFAT samples the design space locally and takes smaller steps towards the optimum.
- DoE explores broadly, but samples inefficiently, leading to wasted experiments farther away from the global optimum.
- Trellis navigates the response landscape efficiently using Bayesian optimization, balancing exploitation and exploration.

Virtual lab setup



What is the virtual lab?

A literature-grounded virtual CHO fed-batch lab that simulates cell growth, metabolism, and product formation as a function of process variables such as temperature feed rate, and temperature. Brute-force simulations identified the ‘true’ maximum titer in the design space.



Why did we build it?

A safe, repeatable sandbox to test models, experimental strategies and active-learning loops before taking them into the wet lab. It enables repeated comparisons at scale and benchmarking against a known underlying process landscape.



How does it work?

Published CHO mechanisms are integrated into coupled dynamic models and parameterised from reported data. For each DoE strategy, this simulator is used for simulating the proposed conditions, running virtual batches, receiving measurements, updating the models and selecting what to test next.